Wiper Control System - High-Level Design

Document version 1.0

Status: released for review

1 Introduction

This High-Level Design (HLD) describes the software architecture for front wiper control with intermittent mode and rain sensor based automatic mode. It lists the software components, their ports and interfaces, the signals exchanged between them and the main functional flows.

The document is the reference for architects, developers, integrators and test engineers. Names of components, interfaces and signals are written exactly as they appear in the software architecture model.

1.1 Scope and assumptions

The design covers the application layer only. Basic software, bootloader and network management are described in separate documents. All components run in the same ECU and communicate through the runtime environment (RTE) using sender-receiver ports.

It is assumed that every signal has exactly one producer. Each signal is refreshed in the cycle time given in the signal list.

1.2 Terms

Term	Meaning
SWC	Software component, the smallest deployable unit of application software.
Port	Connection point of a component. Provided ports send data, required ports receive data.
Interface	Typed description of the data elements exchanged over a port.
Signal	Named data item with one producer and one or more consumers.
RTE	Runtime environment that connects the components.

2 Architecture Overview

The system consists of 4 software components. Each one has a clearly defined responsibility, and the dependencies between components are kept as small as possible. The following sections give the details of every component.

Components that provide data are placed close to the hardware abstraction, while controller components sit above them and make the decisions.

3 Software Components

3.1 Swc_WiperCtrl

Selects the wiper speed from the stalk position and the rain level and requests the speed from the motor driver.

Swc_WiperCtrl has 3 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Rp_StalkPos	required	If_StalkPosition
Rp_RainLevel	required	If_RainLevel
Pp_WiperCmd	provided	If_WiperCmd

Dependencies: Swc_StalkRx, Swc_RainSensorRx.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.2 Swc_StalkRx

Decodes the wiper stalk position from the LIN frame of the steering column module.

Swc_StalkRx has 1 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Pp_StalkPos	provided	If_StalkPosition

Dependencies: none.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.3 Swc_RainSensorRx

Receives the rain intensity from the rain sensor and filters it over 1 s.

Swc_RainSensorRx has 1 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Pp_RainLevel	provided	If_RainLevel

Dependencies: none.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

3.4 Swc_WiperMotorDrv

Controls the wiper motor, detects the park position and reports it.

Swc_WiperMotorDrv has 2 port(s). The ports and the interfaces they use are listed in the table below.

Port	Direction	Interface
Rp_WiperCmd	required	If_WiperCmd
Pp_ParkPos	provided	If_ParkPosition

Dependencies: Swc_WiperCtrl, Swc_WiperDiag.

Error handling: invalid or missing input data leads to a safe default value and the component sets its own status to invalid. The status is available to the diagnostic manager.

4 Interfaces

All interfaces are sender-receiver interfaces. The data elements of each interface are listed below.

Interface	Type	Data elements
If_StalkPosition	Sender-Receiver	StalkState, StalkValid
If_RainLevel	Sender-Receiver	RainIntensity, RainValid
If_WiperCmd	Sender-Receiver	SpeedRequest, ParkRequest
If_ParkPosition	Sender-Receiver	ParkReached

5 Signal List

Producer and consumers of every signal are given in the table. A dash means that no consumer is defined.

Signal	Producer	Consumers	Cycle
Sig_WiperSpeedReq	Swc_WiperCtrl	Swc_WiperMotorDrv	20 ms
Sig_RainIntensity	Swc_RainSensorRx	Swc_WiperCtrl	20 ms
Sig_WiperParked	Swc_WiperMotorDrv	Swc_WiperCtrl	20 ms
Sig_StalkPos	Swc_StalkRx	Swc_WiperCtrl	20 ms

6 Functional Flows

6.1 Automatic wiping

Step 1: Swc_RainSensorRx publishes Sig_RainIntensity.

Step 2: Swc_WiperCtrl maps the intensity to a speed step.

Step 3: Swc_WiperMotorDrv executes the request and reports Sig_WiperParked.

7 Revision History

Version	Date	Change
1.0	2026-04-15	Initial release